

CRYPTOGRAPHIC BILL OF MATERIALS

Post-Quantum Cryptography Readiness Report

Target System	example.com
Scan ID	scan_20260413_183044_3317cb
Scan Date	2026-04-13T13:00:44.431314+00:00
Status	COMPLETED
Generator	Q-Shield CBOM Tool v1.6
Classification	INTERNAL — CONFIDENTIAL

1. Executive Summary

Metric	Value	Status
Total Cryptographic Assets	12	Tracked
Quantum Vulnerable Assets	5	■ ACTION REQUIRED
Quantum Safe Assets	7	✓ COMPLIANT
PQC Readiness Score	0%	Target: 100% by 2035
Overall Risk Severity	UNKNOWN	See Section 3

This CBOM provides a comprehensive inventory of cryptographic assets discovered during automated TLS scanning and readiness for post-quantum migration.

2. TLS Connection Details

Parameter	Value
TLS Version	N/A
Cipher Suite	N/A
Key Exchange	N/A
PQC Ready	No x
Hybrid Key Exch.	No
Crypto Agility	{'score': 89, 'max_score': 100, 'factors': [{ 'name': 'Key algorithm', 'weight': 25, 'score': 100}, { 'name': 'Cipher suite', 'weight': 25, 'score': 89}, { 'name': 'Key exchange', 'weight': 25, 'score': 89}, { 'name': 'Signature', 'weight': 25, 'score': 89}]}

3. Quantum Vulnerability Findings

#	Asset Type	Name	Risk	Description	Recommendation
1	cipher_suite	ECDHE-ECDSA-AES128-GCM-SHA256	HIGH	—	Prefer TLS 1.3 cipher suites with PQC key exchange when available
2	key_exchange	ECDHE	HIGH	—	Enable ML-KEM (Kyber) or hybrid X25519+ML-KEM key exchange
3	cipher_suite	TLS_AES_256_GCM_SHA384	MEDIUM	—	Prefer TLS 1.3 cipher suites with PQC key exchange when available

#	Asset Type	Name	Risk	Description	Recommendation
4	certificate	CN=example.com	HIGH	—	Request PQC-signed certificate from CA when available
5	public_key	ECDSA (secp256r1) 256-bit	HIGH	—	Migrate to ML-DSA (Dilithium) or hybrid ECDSA+ML-DSA certificate

4. Remediation Roadmap

Timeline	Action	Priority
Q2–Q4 2026	Expand hybrid ML-KEM support across TLS 1.3	HIGH
2027	Deploy ML-DSA certificate pilots from internal CA	HIGH
2028–2029	Hybrid ML-KEM + ML-DSA on customer-facing svcs	MEDIUM
2030	Deprecate RSA-only chains; full NIST alignment	HIGH
2031–2035	Retire classical-only cryptography	MEDIUM

5. Audit Trail

Report Generated: 2026-04-13T13:16:35.660902+00:00
Generator: Q-Shield CBOM Assessment Tool v1.6
Compliance: NIST FIPS 203/204/205 · NSA CNSA 2.0
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Disclaimer: This CBOM represents cryptographic assets discovered during automated scanning. Regenerate after infrastructure changes or at least quarterly.